

Birla Institute of Technology & Science, Pilani
1st Semester 2019-2020
Mid-Semester Test (Closed Book)

Course No: **EEE F311**
Maximum Marks: **50**

Course Title: **Communication Systems**
Duration: **90 Minutes**

Date: **03-10-2019**

Write down assumptions and definitions wherever there is an ambiguity.
Please write the final answer and important steps in rectangular boxes.

Q1. $[4 \times 2 = 8 \text{ Marks}]$.

- (a) A sinc pulse exists for infinite duration. However, for visualization, the pulse is shown only for the duration $-4 \leq t \leq 4$ in Figure 1: Q1(a). Find the average energy of the pulse when $-\infty < t < \infty$.

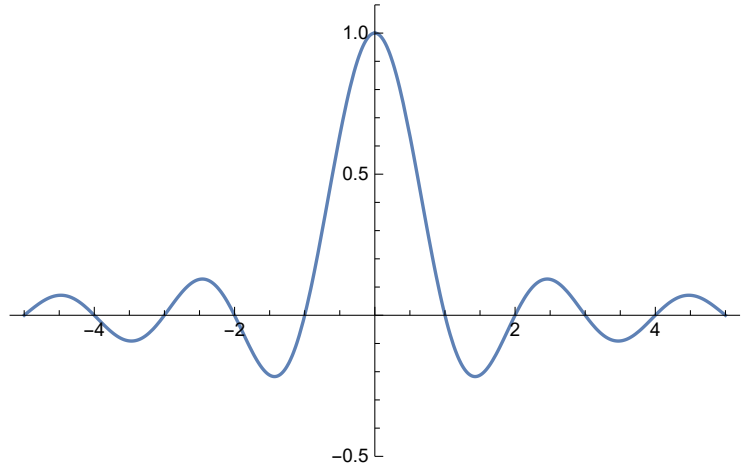


Figure 1: Q1(a)

- (b) The SSB signal $x_{SSB}(t) = m(t) \cos 10000\pi t - m_h(t) \sin 10000\pi t$ is passed through an envelop detector, where the bandwidth of message signal $m(t)$ is 100 Hz. Find the signal at the output of the envelop detector.
- (c) A signal with an average power to the peak power ratio $\frac{1}{1000}$ is encoded using PCM. For the DPCM, the ratio of average power to the peak power of the error signal becomes $\frac{1}{10}$. What will be the cost saving by the DPCM compared to the PCM at the same SQNR of 60 dB. The maximum frequency content of the signal is 20 KHz and transmission of data at 10bps costs 1 Rs. Ignore the additional hardware cost required in implementing the DPCM transmitter and receiver.
- (d) Find the Nyquist interval at the output of sampler (see Figure 2: Q1(d)) if the signal is sampled at the Nyquist rate.

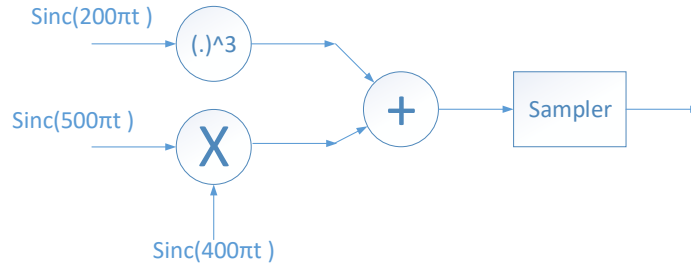


Figure 2: Q1(d)

- Q2. A message signal $m(t)$ has a normalized magnitude (i.e. $|m(t)| = 1$). The message is amplitude modulated using a sinusoidal carrier. The amplitude modulated signal is given as $x(t) = 10 \cos 800\pi t + 40 \cos 1000\pi t + 10 \cos 1200\pi t$. $[4 \times 2 = 8 \text{ Marks}]$
- Find % power efficiency of the modulated signal.
 - Find the modulation index.
 - Find the bandwidth of the modulated signal.
 - Write the phase modulated (PM) signal in time-domain when $k_p = 50\pi$ radian/Volt.
- Q3. A message signal $m(t) = 10 \cos(100\pi t)$ is frequency modulated by a sinusoidal carrier of frequency 1000 Hz and unit amplitude, and transmitted over a channel as shown in Figure 3: Q3. The modulator frequency sensitivity constant is $k_f = 25$ Hz/Volt. $[4 \times 2 = 8 \text{ Marks}]$.



Figure 3: Q3

- Find an approximate bandwidth of the channel required to transmit the modulated signal.
 - Find the time-domain signal at the output of the channel if $h(t) = \frac{1}{5}\delta(t)$.
 - Find the time-domain signal at the output of filter $f(t)$ whose frequency transfer function is $F(f) = j2\pi f$.
 - Find the signal at the output of DC blocker.
- Q4. A sinusoidal signal $m(t) = 1 + \sin 0.25\pi t$ is sampled and fed to a zero-order hold circuit as shown in Figure 4: Q4. $[4 \times 2 = 8 \text{ Marks}]$.
- Find the impulse response of the zero-order hold circuit.
 - Find the transfer function of the zero-order hold circuit.
 - Sketch the sampled signal for one cycle of the sinusoidal signal. Clearly mark the axes.
 - Sketch the output signal $x(t)$ for one cycle of the sinusoidal signal. Clearly mark the axes.

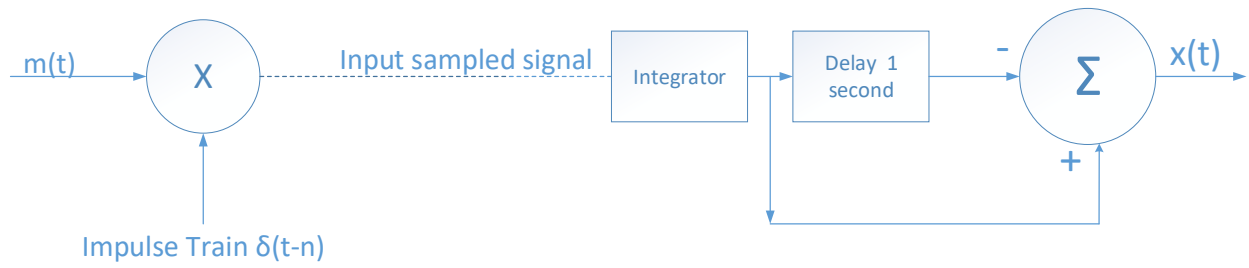


Figure 4: Q4

- Q5. (a) A random bit stream ...101110110100... is encoded using polar line coding (take an ideal rectangular pulse of full pulse width i.e. NRZ). The data is transmitted at a rate 1000 pulses per second over a channel bandwidth 5 KHz. The amplitude of rectangular pulse is +2 volts for binary digit '1' and -2 volts for binary digit '0'. Find the power spectral density of line coded waveform. [4 Marks].
- (b) You have a 2 GigaByte (2GB) per day Internet plan with an average speed of 8 Mbps. The service provider allocates a channel bandwidth 200 KHz for data transmission. Assuming that the signaling is done at the maximum rate, calculate number of data symbols received in downloading a file till the data plan is exhausted. (Use 1GB= 10^9 bytes.) [4 Marks].
- Q6. A periodic message signal $m(t)$ with signal dynamics as shown in Figure 5: Q6 (i) is fed to a PCM system as shown in Figure 5: Q6 (ii). The switch selects either an uniform quantizer without compressor or a compressed system using μ -law. [Marks 4+2+4]

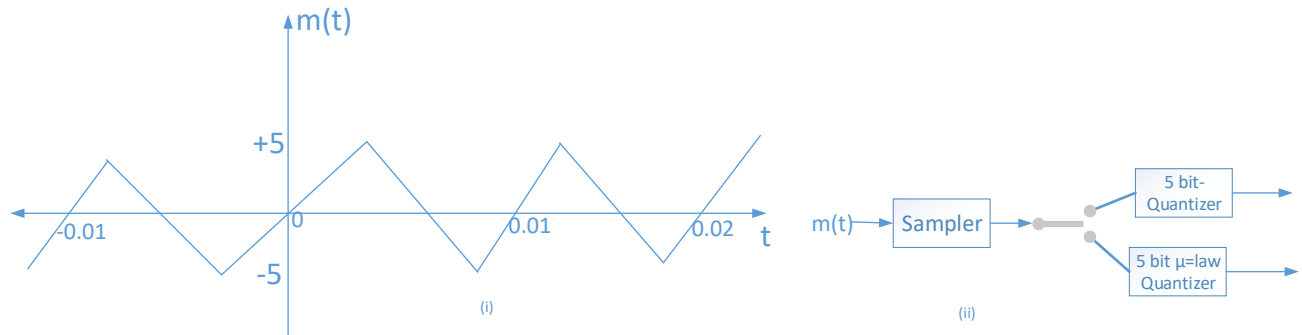


Figure 5: Q6

- (a) Find the SQNR of the signal at the quantizer output if the switch is always ON in the position of non-compression quantizer.
- (b) Find the SQNR of the signal at the quantizer output if the switch is in the ON position of compression quantizer whenever the output SQNR becomes independent of message signal power. Take $\mu = 100$.
- (c) If the message signal $m(t)$ is fed to a DM modulator with a step size 0.1 volts, find the minimum sampling frequency to avoid the slope-overload distortion.

End of Question Paper
